Beyond Frequent Co-Purchases

A Study of Temporal Stability in Retail Association Rules

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AIAA3111 / GROUP 11
Introduction to Data Mining

79.2%
Mean monthly retention
225 of 284 held-out rule-months pass every evaluation gate.

46 / 71
rules pass in all four held-out months

99.3%
of test rule-months are evaluable

01 A test of future reliability

Do associations found in past orders remain reliable in later months?

541,909 raw line items → **19,773** cleaned baskets.
17,901 UK invoices define the primary population.

1. **Discover** — Dec 2010-May 2011
   7,365 orders · Apriori + FP-Growth · length ≤ 3
2. **Select** — Jun-Jul 2011
   2,696 orders · Tune on validation only
3. **Freeze & test** — Aug-Nov 2011
   7,096 orders · 71 fixed rules × four months

Retained = all four gates pass

≥ 0.5% monthly support	**≥ 50%** monthly confidence
≥ 1.2 monthly lift	**≥ 20** antecedent orders

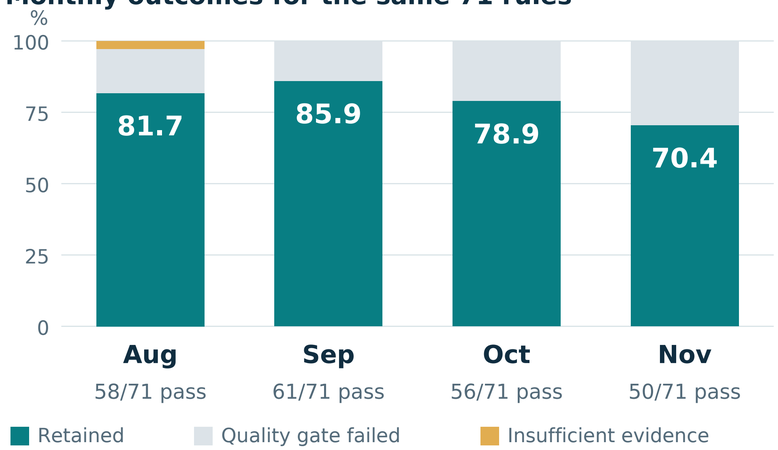
All **71 frozen rules** stay in the denominator. Too little evidence is reported separately from a failed quality threshold.

Audit before mining

Use invoice-level product sets; remove duplicate, invalid and non-product lines and cancellations. Keep singleton and missing-ID baskets. Partial Dec 2011 is excluded from testing.

02 What survives over time?

Monthly outcomes for the same 71 rules



All 71 rules are evaluable in Sep-Nov. The November decline is therefore not a sample-count failure.

Persistence across four test months



Each square is one rule.

Population choice changes the answer

Primary UK invoices	79.2%
Customer ID recorded	64.8%
Baskets ≤ 100 products	62.3%

Mean retention · same 71 rules · descriptive subsets

03 The coverage trade-off

Test retention (%) and discovered rule count

Discovery minimum confidence \ Discovery minimum support	0.5%	1%	2%
30%	**11.7%** 27,943 rules	**37.4%** 1,543 rules	**41.1%** 180 rules
50%	**12.6%** 15,855 rules	**56.7%** 816 rules	**79.2%** 71 rules · chosen
70%	**9.6%** 5,224 rules	**41.6%** 291 rules	**100.0%** 14 rules*

***100% retention, only 14 rules.** This cohort is below the predeclared minimum of 20 rules.

The 71-rule cohort was chosen on **Jun-Jul validation only**. Test-grid comparisons are descriptive.

15,855 rules at 0.5% support / 50% confidence

19.9% evaluable rule-months

12.6% retained rule-months

Equivalent mining outputs

Min. support	Apriori	FP-Growth
0.5%	5.37 s	5.32 s
1%	1.54 s	1.70 s
2%	0.32 s	0.50 s

Median of 3 runs · Apriori low-memory · Ryzen 9 7940H
Same itemsets/counts · peak worker memory < 159 MiB

Frequent co-purchases need temporal validation.
Report retention and observability; choose coverage before examining test results.

One retailer · Four test months · Dependent rules
No claim of causation or sales uplift.

DATA Chen (2015), UCI Online Retail · doi:10.24432/C5BW33 · CC BY 4.0
METHODS Agrawal & Srikant (1994); Han et al. (2000); Raschka (2018), MLxtend.
AIGC Codex assisted code, experiments and drafting; team review and revision remain required.

Code + evidence
Scan the project repository